

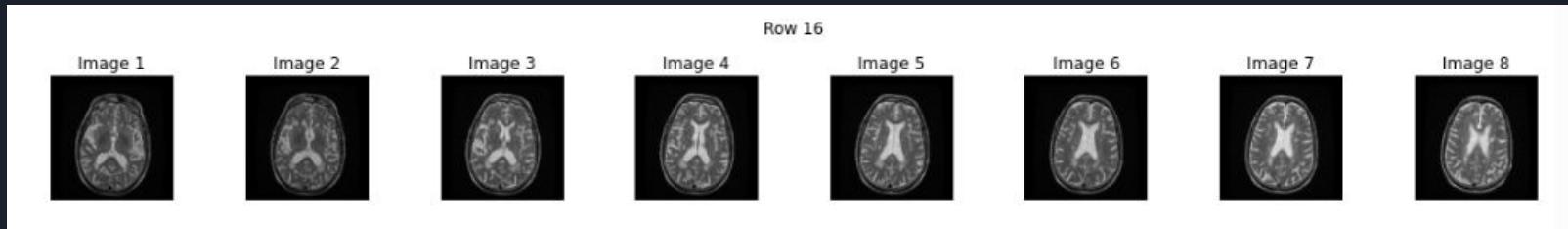
A decorative graphic on the left side of the slide consisting of two overlapping parallelograms. The front one is blue and the back one is a light green. They are positioned diagonally, with the blue one partially covering the green one.

ML for Alzheimer's Classification

By: Jake Treska

Project Specifics

- Utilize the ADNI website to inspect thousands of MRI images
 - 3 Categories: Cognitive Normal (CN), Mild Cognitive Impairment (MCI), Alzheimer's (AD)
- Utilize this data to create a realistic and accurate dataset for training
 - Each MRI is roughly 50 2D slices
 - Certain slices are better at detecting Alzheimer's than others
 - Filter through each MRI image only collecting most important slices
 - Middle 4 images are selected for model training
- Implement Deep Learning frameworks to predict CN,MCI,AD



Model Architecture

- Deep learning model should be able to accurately analyze each MRI slice, in addition to analyzing temporal dimensions (time) between sequential slices.
- Two best models included
 - CRNN (Convolutional Recurrent Neural Network)
 - Inception network

CRNN

Model: "model_1"

Layer (type)	Output Shape	Param #
input_2 (InputLayer)	[(None, 4, 128, 128, 3)]	0
time_distributed (TimeDistri	(None, 4, 128)	2487616
lstm (LSTM)	(None, 4, 32)	20608
dropout_4 (Dropout)	(None, 4, 32)	0
lstm_1 (LSTM)	(None, 16)	3136
dropout_5 (Dropout)	(None, 16)	0
dense_1 (Dense)	(None, 64)	1088
dropout_6 (Dropout)	(None, 64)	0
dense_2 (Dense)	(None, 3)	195

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Total params: 2,512,643
Trainable params: 2,511,683
Non-trainable params: 960

InceptionV3

Model: "sequential"

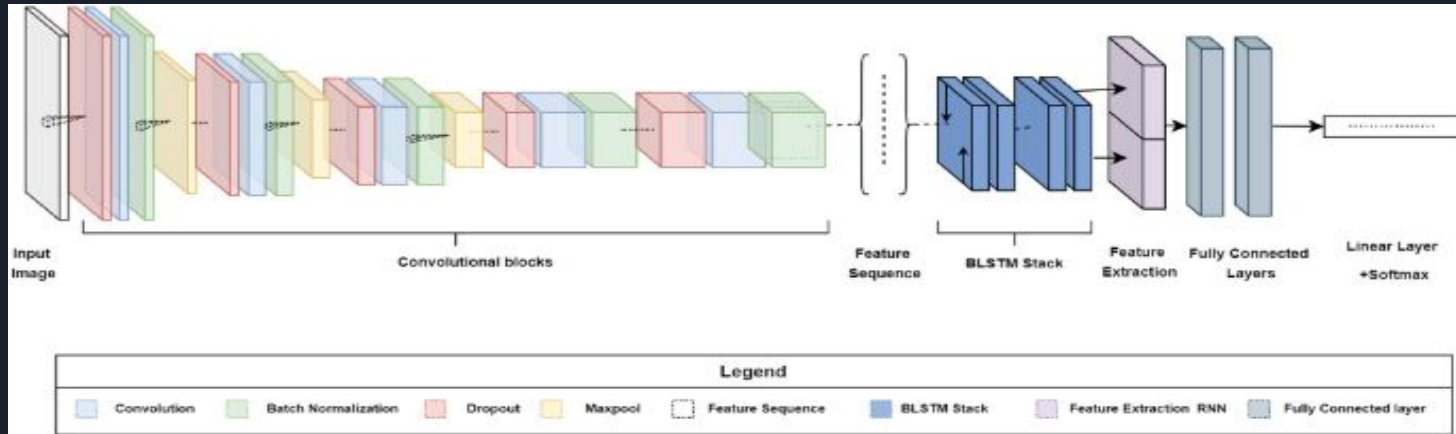
Layer (type)	Output Shape	Param #
time_distributed_1 (TimeDist	(None, 4, 2, 2, 2048)	21802784
time_distributed_2 (TimeDist	(None, 4, 2048)	0
lstm_2 (LSTM)	(None, 32)	266368
dropout_7 (Dropout)	(None, 32)	0
dense_3 (Dense)	(None, 32)	1056
dropout_8 (Dropout)	(None, 32)	0
dense_4 (Dense)	(None, 3)	99

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Total params: 22,070,307
Trainable params: 22,035,875
Non-trainable params: 34,432

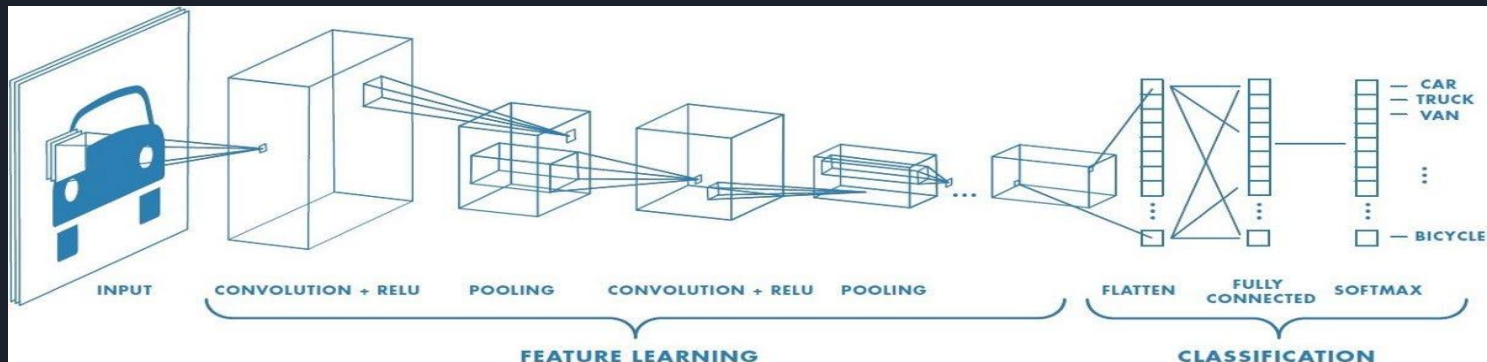
CRNN (Convolutional Recurrent Neural Network)

- CRNN is a CNN (Convolutional Neural Network) combined with a Recurrent Neural Network (RNN).
- CNN is responsible for spatial dimensions (image analysis)
- RNN is responsible for temporal dimensions (time)
- Fully connected layers (FC) used at the end for final prediction
- 3 Total parts = CNN + RNN + FC



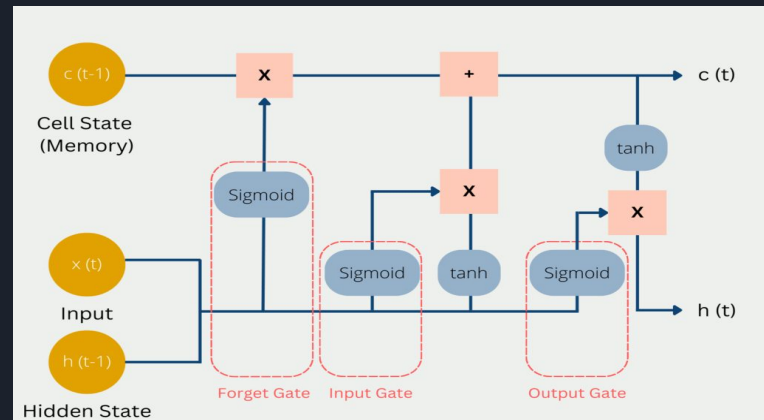
Convolutional Neural Network (CNN)

- Uses convolutional layers to map/highlight important features (shapes/edges)
- Image is split into matrices and multiplied by filter matrix (matrix sizes are customizable)
- Output matrix is then passed through activation functions
- Pooling layers are used to decrease spatial dimensions while keeping important features
- Data is then flattened to 1D array and passed through Dense Layer (FC)
- Dense Layer analyzes extracted features for final predictions



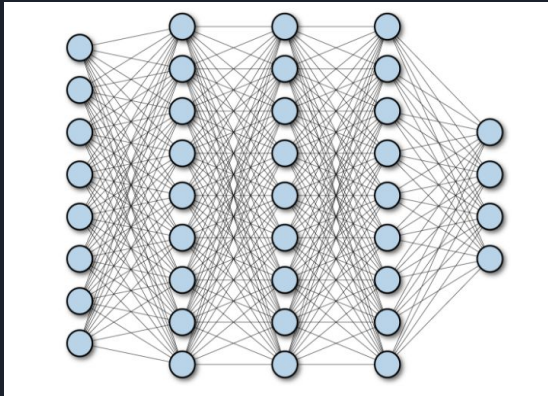
RNN (Recurrent Neural Network)

- Since our model analyzes 4 slices for each MRI image, there will be four 1D vectors after flattening the convolutional layers
- These layers are then passed through a LSTM (Long Short-Term Memory Layer) to extract temporal features
 - LSTM layers are not the only RNN layers, but it is the layer used in this model
- LSTM layers find differences between similar sections across all four 1D vectors, essentially mapping the temporal dimension
- *Cell State*: Total Memory across time steps
- *Input*: New information into LSTM
- *Hidden State*: Output of previous timestep
- *Forget Gate*: Controls what to forget from cell state
- *Input Gate*: Controls what to add to cell state
- *Output Gate*: Controls output, next states hidden state



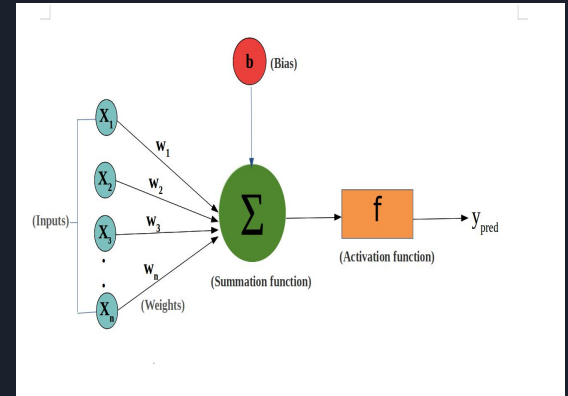
Fully Connected Layer (FC)

- Fully Connected layers, also called Dense Layers, are layers of perceptrons
- Each perceptron is connected to every perceptron in the previous and following layer
- Each perceptron has inputs and a single output
- Inputs are multiplied by a weight value and summed together
- A bias value is added at the very end
- The resultant value is then passed through an activation function
 - ReLU, Tanh, Sigmoid, softmax, etc



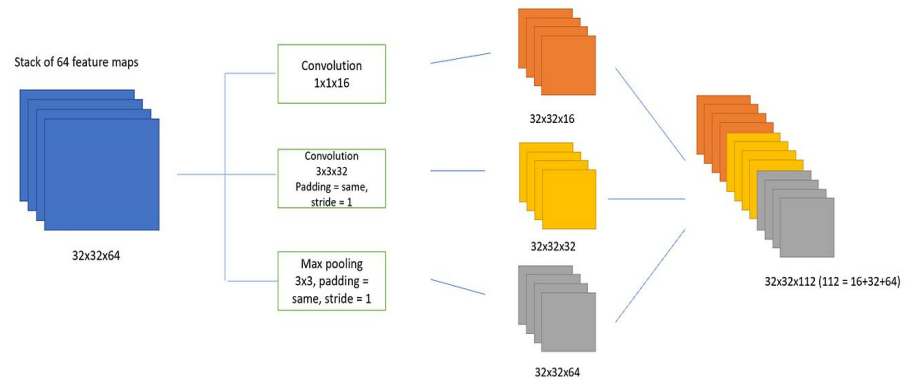
← Fully Connected Layers

Individual Perceptron →



Inception Network

- Inception networks are similar to CNNs
- CNNs apply convolutional layers sequentially
- Inception networks apply convolutional layers in parallel
- Results are concatenated
- Allows the model to better learn across whole training process



Results

- CRNN - 75% Accuracy
- InceptionV3 - 81.7% Accuracy
- *Ensembling*: Combining models for higher prediction
 - (Prediction CRNN + Prediction InceptionV3) / 2
- CRNN + InceptionV3 = 82% Accuracy
- *Precision* = $TP / (TP + FP)$ (Accuracy of how many positive predictions were correct)
- *Recall* = $TP / (TP + FN)$ (Sensitivity towards finding all positive classes)
- *F1 - Score* = $2 \times (Precision \times Recall) / (Precision + Recall)$
- *Support* = Number of Positive Instances

Ensembled Metrics →

Ensemble Accuracy: 0.82					
Classification Report:					
	precision	recall	f1-score	support	
0	0.95	0.82	0.88	51	
1	0.69	0.79	0.73	42	
2	0.82	0.83	0.83	60	
accuracy			0.82	153	
macro avg	0.82	0.81	0.81	153	
weighted avg	0.83	0.82	0.82	153	



Future Plans

- Explore Additional Datasets to allow better training
- Implement tools like Freesurfer to better process/prepare data
- Implement larger models capable of interpreting more MRI slices



Thank You!

I would like to take this time to personally thank my mentor, Dr. Wu, in addition to all the fellow interns and staff. This has been a great experience over the last few months, and I hope we get the chance to collaborate further in the future!